ing the diameter tolerances greater and by expanding the bolts after they have been fastened. It is, however, necessary that the profile of the thread be as nearly correct as possible.

During the last two years 42 boilers have been built which have been equipped with expanded staybolts. None of these staybolts has subsequently been found to be broken or loose. One boiler in the State Railway Repair Shop in Munich has also been fitted with expanded staybolts. This boiler is on a Bavarian locomotive of the type S 3/5; of the 1,360 bolts in this boiler only one bolt has proven to be loose at a pressure of 310 pounds per square inch cold and six of them at the same pressure showed a very slight leak—about one drop of water a minute. At steam pressure and during trial trips all the bolts were found to be perfectly tight. The saving obtained as a result of installing these bolts was 30 percent in comparison with the installation of oversize bolts as previously practiced in this shop. So much for the extract from the article mentioned at the beginning.

PRACTICAL APPLICATION OF STAYBOLT STUDIES

If we now consider the question of the manufacture of staybolt taps of which only those ground in the thread after hardening are considered by Germany's more prominent boiler manufacturers to the limits here suggested, American staybolt tap manufacturers will find that they are lagging behind in obtaining the accuracy in these details as compared with foreign practice. It is a generally conceded fact that staybolt taps to these close limits are actually made in Germany today. Quantity manufacture of this tool, however, is another matter and may be left to the imagination of the reader. American staybolt tap manufacturers will find that there is a real difficulty to solve here and that the problem is of no mean importance.

The total tolerance permitted on these taps for the 4 important elements are, on the outside diameter, the pitch diameter and the root diameter from standard to standard: 0.03 millimeter (0.00118 inch) and on lead 0.01 millimeter (0.00039 inch) in a length of 25 millimeters (approximately 1 inch); the limit on the angle of the thread is 30 feet.

In the case of staybolt taps not ground in the thread after hardening there is, of course, no insurmountable difficulty in holding the taps within the prescribed diameter tolerances with accurate threading tools, correct hardening methods and raw material well suited for the purpose. The lead, however, will invariably be way off from the specified limits for this element. Where staybolt taps are ground in the thread, on the other hand, the tolerances for this element as well as for the outside diameter and the pitch diameter can easily be held within the prescribed limits but the root diameter is more difficult. As far as the writer knows, no thread grinding wheels have yet been developed which can for any length of time maintain an outside diameter thread point and, at the same time, be productive. Present types of thread grinding wheels on the market which can maintain this point for any length of time will besides create great risks of burning or drawing the temper of the threads on the tap while grinding.

As can readily be figured out, the thread points on the grinding wheels can only be permitted to wear down ½ of 0.03 millimeter (0.00118 inch) or 0.0115 millimeter (0.00059 inch) before the root diameter of the tap will be beyond permissible limits. In other words, the thread grinding wheel can only be permitted to wear down on its diameter 0.03 millimeter (0.00118 inch) before it will produce a root diameter in the tap which will pass inspection. With ordinary thread grinding wheels found on the market today such wheels could not take a single cut across the thread length of the tap before they would be worn down sufficiently to result in a non-acceptable root diameter according to the above limits. In Germany the Whitworth form of thread is largely used and for this tap the thread point on the grinding wheel will wear much better than for the U. S. form of thread used in this country due to the fact that its contour is curved as opposed to the flat formation of the U. S. thread.

We will suppose, however, that we have succeeded in making staybolt taps to these so-called necessary limits which, of course, is not an impossibility even if impractical and certainly uneconomical. The next problem is to make these taps cut within the prescribed limits. It will surely not do to make flutes hooked or with a positive rake which for most boiler materials has been found necessary in order to eliminate torn threads in the tap holes and to make the taps cut freely and easily. Staybolt taps so fluted will invariably cut holes larger than the tap diameter. Further, it will not do to give such taps relief in the angle of their thread which has also been found necessary for free and easy cutting because this will also tend to make the taps cut larger than their diameter. The specifications for these taps made in Germany call for 4 radial flutes and with no relief on the straight part of the thread, the 4 flutes instead of the usual 5 being used only in order to facilitate measuring.

The question now arises whether it is actually necessary to make the staybolt taps to such close limits and tolerances as these German experiments and investigations seem to warrant. Offhand, the answer, in the writer's opinion, should be an emphatic *no* because no one can claim that American locomotives and boilers are not as efficient and not as safe as any made elsewhere in the world. For tapping such boilers here unground staybolt taps to very much larger limits are used almost exclusively. Here again staybolt taps that are ground in after hardening are considered wholly unnecessary and as a luxury. Someone might, of course, say that American boilers have a much larger repair bill for upkeep than do German boilers. This is a phase of the subject the writer knows little about. He can only refute such statements with the fact that American boiler and locomotive manufacturers seem to be more than holding their own against all European competition in this line both financially and practically. In other words, they are making money and boilers that are satisfactory for all practical purposes.

Annual Meeting of the American Welding Society

THE eighth annual meeting of the American Welding Society is to be held in the United Engineering Societies Building, 33 West 39th Street, New York City, April 27 to 29 inclusive. Briefly, the program is to include the American Bureau of Welding and Committee meetings on Wednesday, April 27, the former being held in the afternoon and the Gas Welding Committee and Electric Arc Welding Committee meetings to be held in the morning. On Thursday, April 28 the main feature of the day will be a "symposium of research activities in the welding field during the past three years," and a "symposium on production welding." On Friday, April 29, the board of directors will hold its annual meeting in the morning while in the afternoon the committee of the joint pressure vessel committee will hold its hearings.

A. R. CHALKER, chief draftsman of the Locomotive Stoker Company, Pittsburgh, Pa., has been appointed mechanical engineer with headquarters at Pittsburgh, to fill the vacancy caused by the recent death of L. E. Osborne.

staybolt ram, it is the opinion of some boiler makers that this places an undue strain on the wrapper sheet and has a tendency to open the pores of the iron, and that, in districts where the water condition has a pitting tendency, the sheet is more apt to begin pitting in a shorter period of service. On the other hand cutting the bolts with an acetylene torch no doubt has a bad effect on the sheet. However, the method of removing the staybolts is entirely dependent upon each individual railroad and boiler foreman, and their experience should fully cover this matter.

Some roads require that all staybolts be renewed one size larger than the bolt which was removed. This arrangement quickly enlarges the holes in the outside wrapper sheet, and, unless the holes in the wrapper sheet are reduced, the renewal of outside wrapper sheets will be rather heavy.

The proper closing of the holes in the outside wrapper sheet is readily accomplished by using a flat die with a long stroke hammer and holding on the inner side as well as outside, care being taken to only close the holes enough to obtain a full size thread.

The question arises as to the largest size bolt to be applied to a side sheet. There is a difference of opinion on this subject. The smaller the bolt, the more flexible it will be, and as flexibility governs the service given by the staybolts, it is necessary then to keep the size as small as possible. On some roads, when building new boilers, in the first installation staybolts are 7/8 inch diameter. This practice, if within the factor of safety, will lengthen the wrapper sheet service.

Several methods of reducing the size of staybolt holes are in use at the present time; namely, bushings, solid plugs and welding. Bushings are readily made on an automatic screw machine or turret lathe, from solid bars. Where these bushings are made on an automatic machine, it greatly reduces the cost of manufacture. Bushings should be applied in the smallest possible size, so as to allow for renewal, if necessary, when the firebox is next removed.

Solid plugs, or sunflowers, are also used to a great extent. These, as well as the bushings, should be riveted over inside and outside, and great care should be taken in their application to insure tight fit. Applying bushings or solid plugs to crown bolt holes is not practical, owing to the radius of the sheet.

Welding of staybolt holes has its advantages and disadvantages. This system is no doubt the easiest for the boiler maker, as it is only necessary for him to countersink the holes. Tapping a welded hole, especially electrically welded, is very hard on the staybolt taps. Shop conditions should govern the manner of reducing staybolt holes.

Staybolt holes should be drilled 1/8 inch smaller than the required tap size. It is permissible to punch staybolt holes in the outside wrapper sheet. The holes should be punched undersize and reamed to 1/8 inch smaller than the required size for tapping. By reaming the holes, this will give the holes in the outside wrapper sheet the same angle as the holes in the firebox sheet.

The question arises as to the proper kind of taps. Practically all tap manufacturers have their own idea as to the size, amount of taper, and amount of full threads for the taps of their own make. Also, much has been written as to the relative merits of a straight or fluted tap. On a great many stock taps, both straight and spiral fluted, the maximum reamer diameter is so much smaller than the root diameter of the sizing threads, that the threaded section of the tap must do a certain amount of reaming, and, as a result, the cuts are very fine, and, being very light, do not clear themselves, but wad up in the flutes, or gum up between the teeth. The heat generated when this occurs is sufficient to draw the temper of the teeth causing the tap to tear instead of cut.

The most nearly perfect tap will have a maximum reamer diameter of not more than .025 below the root diameter of the sizing threads, the root of the sizing threads to have the same taper as the reaming portion of the tap.

In re-tapping a hole $\frac{1}{16}$ inch larger, the teeth, on the first few revolutions of the taper section, have the same form as the old thread, and have a tendency to act as a lead screw, so that the new thread will follow the old. The remainder of the taper section will cut out part of the old thread, and form a means of steadying the tap in the hole before the new root diameter is reached, and prevent oversize reaming.

A great many railroads have their own specifications for staybolt taps. Others use what is known as the standard stock tap, supplied by the different tap manufacturers.

In selecting taps, the proper procedure is to purchase taps from a number of tap manufacturers and test each make of tap and, from the test made, the tap which gives the best service should be adopted as standard.

In making a tap test, laboratory methods should be followed as nearly as possible. Tapping variables, such as boiler plate hardness and thickness, distance between sheets, size of drilled holes, air pressure, tapping speed, and kind of lubricant, should be strictly adhered to. As far as practical all tests should be made under actual working conditions; that is, a locomotive that is undergoing repairs should be used for the tests. An old sheet, from which staybolts have been removed, will give a more accurate test than two new sheets. Staybolt taps, while on test, should be carefully checked with a ring gage, as to size, and the tapped holes checked with a plug gage, and should be sharpened whenever necessary, and used until the allowable undersize is reached.

Staybolt taps of standard manufacture, unless made to a railroad company's specification, have an allowable oversize of approximately three one-thousandths.

All new staybolt taps, when purchased, should be gaged, and taps of the same size should be marked and used together, so as to have all tapped holes approximately the same size.

When applying staybolts to side sheets, a few holes should be tapped and staybolts applied, so that there will be no possibility of the sheets getting out of alinement while being tapped.

When staybolts are being applied at the same place where threaded, it is better to fit the staybolts to the sheet, rather than to a gage. About one in 50 staybolts threaded should be fitted by the threading machine operator. This will eliminate the possibility of staybolts becoming too tight, due to tap wear.

The question of lubricant used for staybolt tapping and threading rests entirely with the boiler shop foreman. A lubricant should be selected that will give the longest tapping service, cut the cleanest threads, and that will not harden if allowed to stand for some time before the bolts are applied. With certain makes of lubricant, the cuttings will harden on the foundation ring, if allowed to stand for any length of time.

The question arises, "When tapping for crownbolts, should a tap be used with continuous threads the full length of the water space, or a short tap with long leads." If the crownbolts are threaded on a turret lathe, threading both ends at the same time, it is better to use a continuous thread tap. If each end of the bolt is threaded separately, it will make no difference if the short tap is used, as